

From atomic entities to physico-chemical systems

Worksheet answers

5 units · 15 questions each, from recall to challenge

Semester 1 · Academic Year 2026–2027

How to use the answers

- Each question is followed by a full worked answer in a **blue box**. Compare your *method* as well as your result: units, significant figures, justification.
- If a Level 1 or 2 question goes wrong, re-read the matching part of the study guide before moving on; the Level 3 and Challenge questions build on them.

Levels

Level 1 ●

Q1–4: recall a definition, apply one formula.

Level 2 ●●

Q5–8: explain a result or chain two steps.

Level 3 ●●●

Q9–12: multi-step problems, interpreting data.

Challenge ★

Q13–15: synthesis, links between units, going beyond the handout.

Contents

Worksheet	Unit
1	Classical description of the atom
2	Quantum model of the atom
3	Quantum numbers and atomic orbitals
4	Electronic configuration and periodic table
5	Atomic properties and periodicity

Worksheet 1 · Classical description of the atom

Data $e = 1.6 \times 10^{-19} \text{ C}$ $m_e = 9.1094 \times 10^{-31} \text{ kg}$ $m_p = 1.6727 \times 10^{-27} \text{ kg}$ $m_n = 1.6749 \times 10^{-27} \text{ kg}$
 $1 \text{ amu} = 1.6605 \times 10^{-27} \text{ kg}$ $N_A = 6.022 \times 10^{23} / \text{mol}$ $c = 3.00 \times 10^8 \text{ m/s}$ $k = 1/(4\pi\epsilon_0) = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$. Calculator allowed.

Q1. Level 1 • Who discovered what?

Match each scientist to the date and to the discovery or idea: Democritus, Dalton, Thomson, Millikan, Rutherford. Discoveries: (a) the nucleus; (b) the elementary charge; (c) indivisible atoms; (d) the electron; (e) compounds contain fixed ratios of atoms of given masses.

Answer

Democritus, 5th century BC: (c). Dalton, 1808: (e). Thomson, 1897: (d). Millikan, 1909: (b). Rutherford, 1911: (a).

Q2. Level 1 • Counting particles

Give the number of protons, neutrons and electrons of $^{12}_6\text{C}$, $^{14}_6\text{C}$, $^{35}_{17}\text{Cl}$ and $^{56}_{26}\text{Fe}$. Which two are isotopes?

Answer

	$p = Z$	$n = A - Z$	$e = Z$
$^{12}_6\text{C}$	6	6	6
$^{14}_6\text{C}$	6	8	6
$^{35}_{17}\text{Cl}$	17	18	17
$^{56}_{26}\text{Fe}$	26	30	26

$^{12}_6\text{C}$ and $^{14}_6\text{C}$ are isotopes: same $Z = 6$, different A .

Q3. Level 1 • Units of length

Express a typical atomic size, 100 pm, in m, nm and Å. Express 1 fm in pm. What is the typical ratio between the size of an atom and the size of its nucleus?

Answer

$100 \text{ pm} = 100 \times 10^{-12} \text{ m} = 1 \times 10^{-10} \text{ m} = 0.1 \text{ nm} = 1 \text{ Å}$. $1 \text{ fm} = 10^{-15} \text{ m} = 10^{-3} \text{ pm}$. Nucleus 10^{-3} to 10^{-2} pm , so the ratio is $100/10^{-2} = 10^4$ to $100/10^{-3} = 10^5$.

Q4. Level 1 • Cathode rays

In Thomson's tube the beam bends towards the positively charged plate. What does this show? Why was it important that the same deflection was obtained with any gas and any metal?

Answer

Opposite charges attract, so the beam carries a **negative** charge. Since the particles are the same whatever the gas or the metal, they are not specific to one element: they are a component of *every* atom (the electron). Hence the atom is not indivisible.

Q5. Level 2 • Ions and isoelectronic species

Give the number of protons, neutrons and electrons of $^{24}_{12}\text{Mg}^{2+}$, $^{27}_{13}\text{Al}^{3+}$, $^{19}_9\text{F}^-$ and $^{40}_{20}\text{Ca}^{2+}$. Which species are isoelectronic, and with which noble gas?

Answer

	p	n	electrons
Mg^{2+}	12	12	$12 - 2 = 10$
Al^{3+}	13	14	$13 - 3 = 10$
F^-	9	10	$9 + 1 = 10$
Ca^{2+}	20	20	$20 - 2 = 18$

Mg^{2+} , Al^{3+} and F^- are isoelectronic (10 electrons, like neon, $Z = 10$). Ca^{2+} has 18 electrons, like argon.

Q6. Level 2 •• Reading Rutherford's experiment

Give the three observations of the gold-foil experiment and, for each one, what it tells us about the atom. Why could Thomson's "plum pudding" model not explain them?

Answer

(1) Most α -particles pass straight through: the atom is mostly empty. (2) A few are strongly deflected: there is a small region of concentrated positive charge that repels the positive α -particles. (3) About 1 in 10^4 bounces back: this region, the nucleus, is tiny but contains almost all the mass (a light object could not stop a heavy, fast α -particle). In the plum-pudding model the positive charge and the mass are spread over the whole atom; the electric force on the α -particle would stay weak everywhere and no particle would bounce back.

Q7. Level 2 •• How heavy are electrons?

(a) Compute m_p/m_e . (b) What fraction of the mass of a ^{12}C atom (exactly 12 amu) is due to its electrons? Conclude.

Answer

(a) $m_p/m_e = 1.6727 \times 10^{-27} / 9.1094 \times 10^{-31} = 1836$.

(b) $6 m_e / (12 \text{ amu}) = 6 \times 9.1094 \times 10^{-31} / (12 \times 1.6605 \times 10^{-27}) = 2.74 \times 10^{-4}$, i.e. 0.027%. The mass of an atom is almost entirely in its nucleus.

Q8. Level 2 •• A stadium-sized atom

Take an atom of diameter 100 pm and a nucleus of diameter 10^{-3} pm. If the nucleus were enlarged to a marble of diameter 1.0 cm, what would be the diameter of the atom? What fraction of the atom's volume does the nucleus occupy?

Answer

Ratio of diameters: $100/10^{-3} = 10^5$. Atom: $10^5 \times 1.0 \text{ cm} = 10^5 \text{ cm} = 1.0 \text{ km}$. Volumes scale as the cube: $(10^{-5})^3 = 10^{-15}$ of the atom's volume.

Q9. Level 3 ••• Charge-to-mass ratios

(a) Compute e/m for the electron and for the proton. (b) An electron and a proton enter the same electric field with the same speed. Which one is deflected more, and by what factor (the deflection is proportional to q/m)? (c) Why did this make the electron easy to discover with cathode rays?

Answer

(a) Electron: $1.6 \times 10^{-19} / 9.1094 \times 10^{-31} = 1.76 \times 10^{11} \text{ C kg}^{-1}$. Proton: $1.6 \times 10^{-19} / 1.6727 \times 10^{-27} = 9.57 \times 10^7 \text{ C kg}^{-1}$.

(b) Same $|q|$, so the ratio is $m_p/m_e = 1836$: the electron is deflected 1836 times more (in the opposite direction).

(c) The very large e/m_e gives large, easily measured deflections, and showed Thomson that the particle was far lighter than any atom.

Q10. Level 3 ••• From amu to grams per mole

One atom of gold has a mass of 196.967 amu. (a) Express it in kg. (b) How many atoms are there in 1.00 g of gold? (c) How many moles is that? Deduce the molar mass of gold and state the general rule linking amu and g mol^{-1} .

Answer

- (a) $196.967 \times 1.6605 \times 10^{-27} = 3.2706 \times 10^{-25} \text{ kg}$ (Table 1-1 gives 3.2707×10^{-25}).
 (b) $N = 1.00 \times 10^{-3} \text{ kg} / 3.2707 \times 10^{-25} \text{ kg} = 3.06 \times 10^{21} \text{ atoms}$.
 (c) $n = N/N_A = 3.06 \times 10^{21} / 6.022 \times 10^{23} = 5.08 \times 10^{-3} \text{ mol}$, so $M = 1.00 \text{ g} / 5.08 \times 10^{-3} \text{ mol} = 197 \text{ g mol}^{-1}$. Rule: a mass of x amu per atom corresponds to $x \text{ g mol}^{-1}$, because $1 \text{ amu} \times N_A = 1.6605 \times 10^{-27} \times 6.022 \times 10^{23} = 1.000 \times 10^{-3} \text{ kg mol}^{-1}$.

Q11. Level 3 ••• Identify the species

Identify each species (symbol, A , charge). Use a periodic table for the symbols. (a) 26 protons, 30 neutrons, 23 electrons. (b) 17 protons, 20 neutrons, 18 electrons. (c) A neutral atom with $A = 39$ and one more neutron than protons. (d) An ion with 10 electrons, 8 neutrons and charge $2-$.

Answer

- (a) $Z = 26$: iron; $A = 56$; charge $26 - 23 = +3$: ${}^{56}_{26}\text{Fe}^{3+}$.
 (b) $Z = 17$: chlorine; $A = 37$; charge $17 - 18 = -1$: ${}^{37}_{17}\text{Cl}^-$.
 (c) $Z + (Z + 1) = 39$ gives $Z = 19$, potassium: ${}^{39}_{19}\text{K}$ (20 neutrons).
 (d) Charge $2-$ means $Z = 10 - 2 = 8$: oxygen, $A = 8 + 8 = 16$: ${}^{16}_8\text{O}^{2-}$.

Q12. Level 3 ••• Density of nuclear matter

A ${}^{12}\text{C}$ nucleus has a mass of about 12 amu and a radius of about 2.7 fm. (a) Compute its density. (b) What would be the mass of a teaspoon (5 cm^3) of nuclear matter? Compare with water (1000 kg/m^3).

Answer

- (a) $m = 12 \times 1.6605 \times 10^{-27} = 1.99 \times 10^{-26} \text{ kg}$; $V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(2.7 \times 10^{-15})^3 = 8.2 \times 10^{-44} \text{ m}^3$;
 $\rho = m/V = 2.4 \times 10^{17} \text{ kg m}^{-3}$.
 (b) $5 \text{ cm}^3 = 5 \times 10^{-6} \text{ m}^3$, so $m = 2.4 \times 10^{17} \times 5 \times 10^{-6} = 1.2 \times 10^{12} \text{ kg}$ (over a billion tonnes), about 2×10^{14} times denser than water. Ordinary matter is light because atoms are almost empty.

Q13. Challenge ★ Millikan's oil drops

Millikan measured the charges of oil drops: 3.2×10^{-19} , 4.8×10^{-19} , 8.0×10^{-19} , 6.4×10^{-19} and $1.12 \times 10^{-18} \text{ C}$. (a) Show that these values are consistent with a single elementary charge and find it. (b) How many extra electrons does each drop carry? (c) Could a drop carry $2.4 \times 10^{-19} \text{ C}$?

Answer

- (a) Divided by the smallest value, $3.2 \times 10^{-19} \text{ C}$, the charges give 1, 1.5, 2.5, 2 and 3.5: not all whole numbers, so $3.2 \times 10^{-19} \text{ C}$ is not the unit. Half of it, $1.6 \times 10^{-19} \text{ C}$, gives 2, 3, 5, 4 and 7: all whole numbers. The largest charge dividing every value is $e = 1.6 \times 10^{-19} \text{ C}$.
 (b) 2, 3, 5, 4 and 7 electrons.
 (c) No: $2.4 \times 10^{-19} / 1.6 \times 10^{-19} = 1.5$ is not a whole number. Charge is quantised.

Q14. Challenge ★ Where does the missing mass go?

(a) Using the masses in amu of the proton (1.0073), the neutron (1.0087) and the electron ($1/1823$), compute the total mass of the particles that make up a ${}^{12}\text{C}$ atom. (b) Compare with the mass of the atom, 12 amu exactly: compute the mass defect Δm in amu and in kg. (c) The energy that holds the nucleus together is $E = \Delta m c^2$. Compute it per atom and per mole, and compare with a

typical chemical bond energy (about 400 kJ mol^{-1}). Why can chemical reactions not change one element into another?

Answer

- (a) $6 \times 1.0073 + 6 \times 1.0087 + 6/1823 = 6.0438 + 6.0522 + 0.0033 = 12.0993 \text{ amu}$.
 (b) $\Delta m = 12.0993 - 12 = 0.0993 \text{ amu} = 0.0993 \times 1.6605 \times 10^{-27} = 1.65 \times 10^{-28} \text{ kg}$.
 (c) $E = 1.65 \times 10^{-28} \times (3.00 \times 10^8)^2 = 1.48 \times 10^{-11} \text{ J per atom (about 93 MeV)}$; per mole, $\times N_A$: $8.9 \times 10^{12} \text{ J mol}^{-1} = 8.9 \times 10^9 \text{ kJ mol}^{-1}$, about 2×10^7 times a chemical bond. Chemical reactions only rearrange electrons; they involve far too little energy to break or rebuild nuclei, so Z is conserved in chemistry.

Q15. Challenge ★ How small is the gold nucleus?

In Rutherford's experiment, an α -particle (charge $+2e$) with kinetic energy 5.0 MeV ($1 \text{ MeV} = 1.6 \times 10^{-13} \text{ J}$) heads straight for a gold nucleus ($Z = 79$). It stops and turns back at the distance d where all its kinetic energy has become electric potential energy $E_p = k q_1 q_2 / d$. (a) Compute d . (b) What does this tell you about the size of the nucleus? Compare with the size of the gold atom (about 140 pm).

Answer

- (a) $E_k = 5.0 \times 1.6 \times 10^{-13} = 8.0 \times 10^{-13} \text{ J}$. $d = k (2e)(79e) / E_k = 8.99 \times 10^9 \times 2 \times 79 \times (1.6 \times 10^{-19})^2 / 8.0 \times 10^{-13} = 4.5 \times 10^{-14} \text{ m} = 45 \text{ fm}$.
 (b) The α -particle is repelled before touching the nucleus, so the gold nucleus has a radius smaller than $45 \text{ fm} = 0.045 \text{ pm}$ (its actual radius is about 7 fm). That is at least 3000 times smaller than the atom (140 pm), consistent with the 10^{-3} – 10^{-2} pm of the handout.

Worksheet 2 · Quantum model of the atom

Data $c = 3.00 \times 10^8 \text{ m/s}$ $h = 6.62 \times 10^{-34} \text{ J} \cdot \text{s}$ $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$ $N_A = 6.022 \times 10^{23} / \text{mol}$
 $R_H = 3.29 \times 10^{15} \text{ Hz}$ hydrogen: $E_n = -13.6/n^2 \text{ eV}$; hydrogen-like ion: $E_n = -13.6 Z^2/n^2 \text{ eV}$. Calculator allowed.

Q1. Level 1 • Ordering the spectrum

Put these radiations in order of increasing frequency: visible light, X-rays, radio waves, infrared, γ rays, ultraviolet, microwaves. Which one has the longest wavelength? Which photons carry the most energy?

Answer

Radio waves < microwaves < infrared < visible < ultraviolet < X-rays < γ rays. Radio waves have the longest wavelength ($\lambda = c/\nu$); γ photons carry the most energy ($E = h\nu$).

Q2. Level 1 • A red laser

A laser emits red light at $\lambda = 650 \text{ nm}$. Compute its frequency ν and its period T .

Answer

$$\nu = c/\lambda = 3.00 \times 10^8 / 650 \times 10^{-9} = 4.62 \times 10^{14} \text{ Hz}; \quad T = 1/\nu = \lambda/c = 2.17 \times 10^{-15} \text{ s}.$$

Q3. Level 1 • Energy of one photon

Compute the energy of one photon of the laser of Q2, in joules and in electronvolts.

Answer

$$E = hc/\lambda = 6.62 \times 10^{-34} \times 3.00 \times 10^8 / 650 \times 10^{-9} = 3.06 \times 10^{-19} \text{ J}; \quad E = 3.06 \times 10^{-19} / 1.6 \times 10^{-19} = 1.91 \text{ eV}.$$

Q4. Level 1 • Vocabulary of the Bohr model

Define ground state, excited state and ionised state for the hydrogen atom, and give the energy of the electron in each case (for the excited state, give E_2 and E_3).

Answer

Ground state: $n = 1$, lowest energy, $E_1 = -13.6 \text{ eV}$. Excited state: any $n > 1$, higher energy, e.g. $E_2 = -13.6/4 = -3.40 \text{ eV}$, $E_3 = -13.6/9 = -1.51 \text{ eV}$. Ionised state: $n \rightarrow \infty$, $E_\infty = 0$, the electron is no longer bound to the proton.

Q5. Level 2 • Emission versus absorption

Describe the emission spectrum and the absorption spectrum of hydrogen gas in the visible. Explain why the lines appear at the same wavelengths in both.

Answer

Emission (gas excited by a discharge): four coloured lines on a black background at 656, 486, 434 and 410 nm. Absorption (white light through the gas): a continuous rainbow with four black lines at the same wavelengths. Both involve the same pairs of levels: in emission an electron falls from E_p to E_n and emits a photon $h\nu = E_p - E_n$; in absorption a photon of the same energy lifts it from E_n to E_p . Same energy, same wavelength.

Q6. Level 2 •• Why is permanganate purple?

An aqueous solution of KMnO_4 absorbs light between about 515 and 565 nm. (a) Which colour is absorbed? What colour does the solution appear? (b) Why is the absorption a broad band and not thin lines as for hydrogen gas?

Answer

(a) 515–565 nm is green. The eye receives the rest of the white light, mainly red and blue–violet, which looks purple/magenta: the complementary colour of green.
 (b) The absorbing species is a molecular ion in a liquid: it has very many closely spaced energy levels (vibrations, interactions with the solvent), so it absorbs a continuous range of energies. An isolated atom in a gas has only a few well-separated levels, hence thin lines.

Q7. Level 2 •• The Balmer lines from Rydberg's formula

Using $\nu = R_H(1/n_1^2 - 1/n_2^2)$ with $n_1 = 2$, compute the wavelengths for $n_2 = 3, 4, 5, 6$ and compare them with the handout values.

Answer

$\lambda = c/\nu$: $n_2 = 3$: $\nu = 3.29 \times 10^{15}(\frac{1}{4} - \frac{1}{9}) = 4.57 \times 10^{14}$ Hz, $\lambda = 657$ nm; $n_2 = 4$: 6.17×10^{14} Hz, 486 nm; $n_2 = 5$: 6.91×10^{14} Hz, 434 nm; $n_2 = 6$: 7.31×10^{14} Hz, 410 nm. They agree with 656, 486, 434, 410 nm (the small gap for the first line comes from rounding c and R_H).

Q8. Level 2 •• Rydberg meets Bohr

(a) Using Bohr's levels, compute the energy (in eV and J) of the photon emitted in the transition $3 \rightarrow 2$, and its wavelength. (b) Show that Bohr's formula gives Rydberg's formula and that $R_H = 13.6 \text{ eV}/h$. Check the numerical value.

Answer

(a) $\Delta E = E_3 - E_2 = 13.6(\frac{1}{4} - \frac{1}{9}) = 1.89 \text{ eV} = 3.02 \times 10^{-19} \text{ J}$; $\lambda = hc/\Delta E = 657 \text{ nm}$ (the red line).
 (b) $h\nu = E_{n_2} - E_{n_1} = 13.6 \text{ eV}(1/n_1^2 - 1/n_2^2)$, so $\nu = (13.6 \text{ eV}/h)(1/n_1^2 - 1/n_2^2)$: Rydberg's formula with $R_H = 13.6 \times 1.6 \times 10^{-19}/6.62 \times 10^{-34} = 3.29 \times 10^{15} \text{ Hz}$. ✓

Q9. Level 3 ••• Series and their limits

(a) Compute the wavelength of the first Lyman line ($2 \rightarrow 1$) and the Lyman series limit. In which region are they? (b) Compute the Paschen series limit ($n_1 = 3$). (c) Explain why only the Balmer series has lines in the visible.

Answer

(a) $2 \rightarrow 1$: $\nu = R_H \times \frac{3}{4} = 2.47 \times 10^{15} \text{ Hz}$, $\lambda = 122 \text{ nm}$. Limit: $\nu = R_H$, $\lambda = c/R_H = 91.2 \text{ nm}$. Both UV.
 (b) $\nu = R_H/9 = 3.66 \times 10^{14} \text{ Hz}$, $\lambda = 821 \text{ nm}$ (IR). All Paschen lines are beyond 821 nm.
 (c) Lyman gaps are at least $E_2 - E_1 = 10.2 \text{ eV}$ (UV); Paschen gaps are at most 1.51 eV ($\lambda \geq 821 \text{ nm}$, IR); Balmer gaps go from 1.89 eV (656 nm) to 3.40 eV (365 nm), which covers the visible range 400–700 nm.

Q10. Level 3 ••• Ionisation energy of hydrogen

(a) What energy is needed to ionise a hydrogen atom in its ground state? Give it in eV, in J per atom and in kJ mol^{-1} . (b) What is the longest wavelength that can do it? (c) Unit 5 lists ionisation energies in kJ mol^{-1} but leaves hydrogen blank: fill it in.

Answer

- (a) $E_\infty - E_1 = 13.6 \text{ eV} = 13.6 \times 1.6 \times 10^{-19} = 2.18 \times 10^{-18} \text{ J per atom}$; $\times N_A$: $1.31 \times 10^6 \text{ J mol}^{-1} = 1310 \text{ kJ mol}^{-1}$.
 (b) $\lambda = hc/E = 91.2 \text{ nm}$ (the Lyman limit, UV); shorter wavelengths ionise too.
 (c) Hydrogen: about 1310 kJ mol^{-1} (the accepted value is 1312).

Q11. Level 3 ••• Counting emission lines

Hydrogen atoms are excited to $n = 4$ and then return to the ground state by all possible paths. (a) How many different lines can be emitted? List the transitions. (b) Sort them into series and spectral regions. (c) Which ones can you see?

Answer

- (a) Any pair of levels among $\{1, 2, 3, 4\}$: $\binom{4}{2} = 6$ lines: $4 \rightarrow 3$, $4 \rightarrow 2$, $4 \rightarrow 1$, $3 \rightarrow 2$, $3 \rightarrow 1$, $2 \rightarrow 1$.
 (b) Lyman (to 1, UV): $4 \rightarrow 1$ (97 nm), $3 \rightarrow 1$ (103 nm), $2 \rightarrow 1$ (122 nm). Balmer (to 2): $4 \rightarrow 2$ (486 nm), $3 \rightarrow 2$ (656 nm), both visible. Paschen (to 3): $4 \rightarrow 3$ (1876 nm, IR).
 (c) Only $4 \rightarrow 2$ (blue-green) and $3 \rightarrow 2$ (red).

Q12. Level 3 ••• Which photons are absorbed?

Hydrogen atoms in their ground state are hit by photons of energy (a) 10.2 eV, (b) 11.0 eV, (c) 12.09 eV, (d) 15.0 eV. For each, say whether it is absorbed and what happens to the atom.

Answer

- The possible gaps from $n = 1$ are $E_p - E_1 = 13.6(1 - 1/p^2)$: 10.20 ($p = 2$), 12.09 ($p = 3$), 12.75 ($p = 4$) ... up to 13.6 eV (ionisation).
 (a) Absorbed, $n = 1 \rightarrow 2$. (b) Not absorbed: 11.0 eV matches no gap and is below 13.6 eV. (c) Absorbed, $n = 1 \rightarrow 3$. (d) Absorbed: $15.0 > 13.6 \text{ eV}$, the atom is ionised and the electron leaves with kinetic energy $15.0 - 13.6 = 1.4 \text{ eV}$.

Q13. Challenge ★ Identify the transition

A line of the hydrogen emission spectrum is measured at $\lambda = 1876 \text{ nm}$. Find n_1 and n_2 . Explain your method.

Answer

$\nu = c/\lambda = 3.00 \times 10^8 / 1876 \times 10^{-9} = 1.599 \times 10^{14} \text{ Hz}$, so $1/n_1^2 - 1/n_2^2 = \nu/R_H = 0.0486$. The line is in the IR, and the Balmer limit is 365 nm, so $n_1 \geq 3$. Try $n_1 = 3$: $1/n_2^2 = 1/9 - 0.0486 = 0.0625 = 1/16$, so $n_2 = 4$. The line is $4 \rightarrow 3$, the first Paschen line. (With $n_1 = 4$, $1/n_2^2 = 0.0625 - 0.0486 = 0.0139$ gives $n_2 = 8.5$, not an integer.)

Q14. Challenge ★ The helium ion

He^+ ($Z = 2$) has a single electron. (a) Give its energy levels E_1 , E_2 , E_3 and its ionisation energy. (b) Compute the wavelength of its $3 \rightarrow 2$ transition; in which region is it? (c) Show that the He^+ transition $6 \rightarrow 4$ has exactly the same wavelength as the red line of hydrogen. Why is this so?

Answer

- (a) $E_n = -13.6 \times 4/n^2 = -54.4/n^2 \text{ eV}$: $E_1 = -54.4$, $E_2 = -13.6$, $E_3 = -6.04 \text{ eV}$. Ionisation energy 54.4 eV.
 (b) $\Delta E = 54.4(\frac{1}{4} - \frac{1}{9}) = 7.56 \text{ eV}$; $\lambda = hc/\Delta E = 164 \text{ nm}$ (UV).
 (c) $6 \rightarrow 4$: $\Delta E = 54.4(\frac{1}{16} - \frac{1}{36}) = 13.6 \times 4 \times \frac{5}{144} = 13.6 \times \frac{5}{36} = 13.6(\frac{1}{4} - \frac{1}{9})$, the same energy as H $3 \rightarrow 2$, so $\lambda = 656 \text{ nm}$. Because $Z^2/n^2 = (Z/n)^2$ and $4/6 = 2/3$, $2/4 = 1/2$: every He^+ level with even $n = 2k$ has the same energy as the H level k .

Q15. Challenge ★ **Photons in a laser beam**

A green laser pointer emits 1.0 mW at 532 nm. (a) How many photons does it emit per second? (b) How many moles of photons per hour? (c) Why do we not notice that light comes in packets?

Answer

- (a) One photon: $E = hc/\lambda = 6.62 \times 10^{-34} \times 3.00 \times 10^8 / 532 \times 10^{-9} = 3.73 \times 10^{-19}$ J. Per second: $1.0 \times 10^{-3} / 3.73 \times 10^{-19} = 2.7 \times 10^{15}$ photons.
- (b) $2.7 \times 10^{15} \times 3600 / 6.022 \times 10^{23} = 1.6 \times 10^{-5}$ mol per hour.
- (c) Each packet is tiny ($\sim 10^{-19}$ J) and there are $\sim 10^{15}$ of them per second: the beam looks continuous, just as water looks continuous although it is made of molecules.

Worksheet 3 · Quantum numbers and atomic orbitals

Data $c = 3.00 \times 10^8 \text{ m/s}$ $h = 6.62 \times 10^{-34} \text{ J} \cdot \text{s}$ $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$ $a_0 = 52.9 \text{ pm}$ hydrogen-like species: $E_n = -13.6 Z^2/n^2 \text{ eV}$, $r = n^2 a_0/Z$. Calculator allowed.

Q1. Level 1 • The third shell

For $n = 3$, list the allowed values of ℓ with the name of each subshell, and the values of m_ℓ for each. How many orbitals does the shell contain?

Answer

$\ell = 0$ (3s): $m_\ell = 0$. $\ell = 1$ (3p): $m_\ell = -1, 0, +1$. $\ell = 2$ (3d): $m_\ell = -2, -1, 0, +1, +2$. Total $1 + 3 + 5 = 9 = 3^2$ orbitals.

Q2. Level 1 • Allowed or not?

Which of these sets (n, ℓ, m_ℓ) are allowed? Name the orbital when it is, and give the reason when it is not: (a) (2, 2, 0); (b) (3, 1, -1); (c) (1, 0, 1); (d) (4, 3, -2); (e) (2, 1, 2); (f) (5, 0, 0).

Answer

(a) No: $\ell \leq n - 1 = 1$. (b) Yes, 3p. (c) No: $\ell = 0$ forces $m_\ell = 0$. (d) Yes, 4f. (e) No: $|m_\ell| \leq \ell = 1$. (f) Yes, 5s.

Q3. Level 1 • Naming wavefunctions

Name the orbital described by $\Psi_{2,0,0}$, $\Psi_{3,1,1}$, $\Psi_{4,2,-2}$ and $\Psi_{4,3,0}$. Conversely, give all the wavefunctions of the 2p subshell.

Answer

2s, 3p, 4d, 4f. 2p: $\Psi_{2,1,-1}$, $\Psi_{2,1,0}$, $\Psi_{2,1,1}$.

Q4. Level 1 • Spin and boundary surfaces

(a) Which values can m_s take, and how are they drawn? Which experiment revealed spin? (b) What is a boundary surface? What is the shape of the boundary surface of an s orbital, and of a p orbital?

Answer

(a) $m_s = +\frac{1}{2}$ ($\uparrow$) or $-\frac{1}{2}$ ($\downarrow$). The splitting of the sodium D line into two lines (589.0 and 589.6 nm).
(b) A surface enclosing a given probability (usually 95 %) of finding the electron. s: a sphere centred on the nucleus; p: two lobes on either side of the nucleus along one axis (dumbbell), separated by a nodal plane.

Q5. Level 2 • Energies and degeneracy in hydrogen

(a) Give the energy of the electron of hydrogen described by $\Psi_{3,2,-1}$ and by $\Psi_{3,0,0}$. What word describes this situation? (b) How many degenerate orbitals are there in the shell $n = 4$ of hydrogen?

Answer

(a) Both have $n = 3$: $E_3 = -13.6/9 = -1.51 \text{ eV}$. In hydrogen the energy depends only on n : the 3d and 3s orbitals are **degenerate**.
(b) $n^2 = 16$ (one 4s, three 4p, five 4d, seven 4f).

Q6. Level 2 •• Ground or excited?

For the hydrogen atom, say which orbital is occupied, whether the atom is in its ground state or an excited state, and its energy, when the electron is described by (a) $\Psi_{1,0,0}$, (b) $\Psi_{2,1,0}$, (c) $\Psi_{3,2,1}$. (d) How much energy is needed to go from (a) to (b)?

Answer

(a) 1s, ground state, -13.6 eV. (b) 2p, excited, -3.40 eV. (c) 3d, excited, -1.51 eV.
(d) $-3.40 - (-13.6) = 10.2$ eV (a UV photon of 122 nm).

Q7. Level 2 •• Counting nodes

Give the number of radial nodes, angular nodes and total nodes of the 2s, 3p, 3d, 4s and 4f orbitals. Check the 3s, 3p and 3d cases on Figure 3-3 of the handout.

Answer

Radial $n - \ell - 1$, angular ℓ , total $n - 1$: 2s: 1, 0, 1; 3p: 1, 1, 2; 3d: 0, 2, 2; 4s: 3, 0, 3; 4f: 0, 3, 3.

On Figure 3-3, the radial curve of 3s falls to zero twice before its main maximum (2 radial nodes), 3p once (1) and 3d never (0).

Q8. Level 2 •• The three p orbitals

For each of p_x , p_y and p_z , give the axis along which it points and its nodal plane. Which one corresponds to $m_\ell = 0$? What happens to the sign of the wavefunction from one lobe to the other?

Answer

p_x : along x , nodal plane (yOz). p_y : along y , nodal plane (xOz). p_z : along z , nodal plane (xOy). p_z corresponds to $m_\ell = 0$ (p_x and p_y are combinations of $m_\ell = \pm 1$). The wavefunction changes sign across the nodal plane: the two lobes have opposite signs (opposite colours).

Q9. Level 3 ••• Size of orbitals

Using $r = n^2 a_0 / Z$, compute the radius of: (a) the 1s, 2s and 3s orbitals of hydrogen; (b) the 1s orbital of He^+ and of Li^{2+} . (c) Explain in one sentence the effect of n and of Z .

Answer

(a) H: 1s: 52.9 pm; 2s: $4 \times 52.9 = 212$ pm; 3s: $9 \times 52.9 = 476$ pm.

(b) He^+ ($Z = 2$): $52.9/2 = 26.5$ pm; Li^{2+} ($Z = 3$): $52.9/3 = 17.6$ pm.

(c) The orbital grows as n^2 (the electron is less bound and further out) and shrinks as $1/Z$ (a larger nuclear charge pulls the electron closer).

Q10. Level 3 ••• Ordering subshells

Using Madelung's rule, write $n + \ell$ for each subshell and put them in order of increasing energy: 5s, 4f, 3d, 4p, 6s, 4d, 5p, 4s.

Answer

$n + \ell$: 4s: 4; 3d: 5; 4p: 5; 5s: 5; 4d: 6; 5p: 6; 6s: 6; 4f: 7.

Order (equal $n + \ell$ sorted by increasing n): $4s < 3d < 4p < 5s < 4d < 5p < 6s < 4f$.

Q11. Level 3 ••• Hydrogen versus potassium

In hydrogen the 3d orbitals have the same energy as 3s, and lie far below 4s. In potassium ($Z = 19$) the 4s orbital is lower than 3d. (a) Which approximation lets us still use the labels 1s, 2s, 2p ... for many-electron atoms? (b) Explain why the energy now depends on ℓ and why 4s can drop below 3d.

Answer

- (a) The orbital approximation: electron–electron repulsions are neglected in front of the attraction of the nucleus, so each electron occupies its own hydrogen-like AO.
- (b) The inner electrons screen the nucleus. An s electron penetrates close to the nucleus (small inner maxima of its radial density), feels a larger effective charge and is more strongly bound than p and d electrons of the same shell: E now depends on ℓ . The 4s orbital penetrates so much that its energy falls below that of 3d, which stays outside the core; Madelung's rule ($n + \ell = 4$ for 4s, 5 for 3d) summarises this.

Q12. Level 3 ••• The Zeeman effect

- (a) Into how many sublevels does a p subshell split in a magnetic field? a d subshell? an f subshell?
- (b) Hydrogen atoms emit the $2p \rightarrow 1s$ line (122 nm). How many lines are seen in a magnetic field?
- (c) What does the Zeeman effect prove?

Answer

- (a) $2\ell + 1$ sublevels: p: 3, d: 5, f: 7.
- (b) 1s does not split ($\ell = 0$); 2p gives 3 sublevels, so 3 lines very close to 122 nm.
- (c) That the orbitals of a subshell have different orientations, labelled by m_ℓ with $2\ell + 1$ values; without a field they are degenerate.

Q13. Challenge ★ How many electrons can a shell hold?

An electron is described by four quantum numbers (n, ℓ, m_ℓ, m_s) . (a) List all the allowed sets for $n = 2$. (b) Show that the number of sets for a shell n is $2n^2$ and check it for $n = 1, 2, 3$. (Unit 4: no two electrons of an atom can share the same set, so this is the maximum number of electrons of the shell.)

Answer

- (a) $(2, 0, 0, \pm\frac{1}{2})$; $(2, 1, -1, \pm\frac{1}{2})$; $(2, 1, 0, \pm\frac{1}{2})$; $(2, 1, 1, \pm\frac{1}{2})$: 8 sets.
- (b) Number of orbitals: $\sum_{\ell=0}^{n-1} (2\ell + 1) = 2 \cdot \frac{(n-1)n}{2} + n = n^2$; each orbital has two values of m_s : $2n^2$. $n = 1$: 2; $n = 2$: 8; $n = 3$: 18.

Q14. Challenge ★ The sodium doublet

The D line of sodium is a doublet: $\lambda_1 = 589.0$ nm and $\lambda_2 = 589.6$ nm. (a) Compute the energy of each photon in J and in eV (4 significant figures). (b) Compute the energy difference between the two upper levels in J and in meV. (c) Explain the origin of the doublet.

Answer

- (a) $E_1 = hc/\lambda_1 = 6.62 \times 10^{-34} \times 3.00 \times 10^8 / 589.0 \times 10^{-9} = 3.372 \times 10^{-19}$ J = 2.107 eV; $E_2 = 3.368 \times 10^{-19}$ J = 2.105 eV.
- (b) $\Delta E = 3.43 \times 10^{-22}$ J = 2.1×10^{-3} eV = 2.1 meV (about 0.1 % of E). Keep all digits until the subtraction: the difference of two close numbers loses precision.
- (c) The line is the transition $3p \rightarrow 3s$ of the outer electron. Its spin ($m_s = \pm\frac{1}{2}$) interacts with its orbital motion, which splits 3p into two levels 2.1 meV apart; 3s ($\ell = 0$) is not split, so there are two lines.

Q15. Challenge ★ Where is the 1s electron?

The radial probability density of the 1s orbital of hydrogen is $P(r) = \frac{4}{a_0^3} r^2 e^{-2r/a_0}$. (a) Show that P is maximum at $r = a_0$. (b) Explain why the probability density is zero at $r = 0$ even though $|\Psi_{1s}|^2$ is largest at the nucleus. (c) For He^+ , $P(r) \propto r^2 e^{-2Zr/a_0}$: where is its maximum?

Answer

- (a) $\frac{dP}{dr} = \frac{4}{a_0^3} e^{-2r/a_0} \left(2r - \frac{2r^2}{a_0} \right) = \frac{8r}{a_0^3} e^{-2r/a_0} \left(1 - \frac{r}{a_0} \right)$, which is zero for $r = a_0$ (positive before, negative after): maximum at $r = a_0 = 52.9$ pm, the Bohr radius.
- (b) $P(r)$ is the probability per unit distance of being in a thin shell of radius r , whose volume $4\pi r^2 dr$ tends to zero at the centre. The density per unit volume is largest at the nucleus, but there is almost no volume there.
- (c) Same calculation with a_0/Z instead of a_0 : maximum at $r = a_0/Z = 26.5$ pm, as given by $r = n^2 a_0 / Z$.

Worksheet 4 · Electronic configuration and periodic table

Data Atomic numbers: H 1, He 2, Li 3, Be 4, B 5, C 6, N 7, O 8, F 9, Ne 10, Na 11, Mg 12, Al 13, Si 14, P 15, S 16, Cl 17, Ar 18, K 19, Ca 20, Sc 21, Ti 22, V 23, Cr 24, Mn 25, Fe 26, Co 27, Ni 28, Cu 29, Zn 30, Ga 31, Ge 32, As 33, Se 34, Br 35, Kr 36, Rb 37, Sr 38. No calculator needed.

Q1. Level 1 • Capacity of subshells

Complete: an s subshell has ... orbital(s) and holds at most ... electrons; p: ... and ...; d: ... and ...; f: ... and Which principle limits an orbital to two electrons?

Answer

s: 1 and 2; p: 3 and 6; d: 5 and 10; f: 7 and 14. The Pauli exclusion principle: two electrons of an orbital share n, ℓ, m_ℓ , so they must differ by m_s ($+\frac{1}{2}$ and $-\frac{1}{2}$).

Q2. Level 1 • The three rules

State the Aufbau principle, the Pauli exclusion principle and Hund's rule in one sentence each.

Answer

Aufbau: in the ground state, electrons fill the lowest-energy orbitals first (Madelung order). Pauli: two electrons of an atom cannot have the same four quantum numbers (at most two per orbital, with opposite spins). Hund: in a subshell, electrons occupy separate orbitals with parallel spins before pairing.

Q3. Level 1 • Writing configurations

Write the full ground-state configuration of O, Na, Cl and Ar, then the shorthand form with a noble gas core.

Answer

O: $1s^2 2s^2 2p^4 = [\text{He}] 2s^2 2p^4$. Na: $1s^2 2s^2 2p^6 3s^1 = [\text{Ne}] 3s^1$. Cl: $1s^2 2s^2 2p^6 3s^2 3p^5 = [\text{Ne}] 3s^2 3p^5$. Ar: $1s^2 2s^2 2p^6 3s^2 3p^6 = [\text{Ne}] 3s^2 3p^6$.

Q4. Level 1 • Core and valence

Give the number of core electrons and of valence electrons of C, Mg, S and Ar.

Answer

C $[\text{He}] 2s^2 2p^2$: 2 core, 4 valence. Mg $[\text{Ne}] 3s^2$: 10 core, 2 valence. S $[\text{Ne}] 3s^2 3p^4$: 10 core, 6 valence. Ar $[\text{Ne}] 3s^2 3p^6$: 10 core, 8 valence.

Q5. Level 2 • Box diagrams and Hund's rule

(a) Draw the box diagram of the valence shell of N, O and F and give the number of unpaired electrons. (b) Which of these $2p^3$ diagrams respects Hund's rule? Explain what is wrong with the others: (i) $\uparrow\downarrow\uparrow\uparrow$ (ii) $\uparrow\uparrow\uparrow\uparrow$ (iii) $\uparrow\downarrow\uparrow\uparrow$.

Answer

(a) N: $\uparrow\downarrow\uparrow\uparrow\uparrow$, 3 unpaired. O: $\uparrow\downarrow\uparrow\uparrow\uparrow$, 2 unpaired. F: $\uparrow\downarrow\uparrow\downarrow\uparrow\uparrow$, 1 unpaired.

(b) (ii). In (i) two electrons pair up while an orbital is still empty; in (iii) the spins are not all parallel. Both are allowed by Pauli but are excited states.

Q6. Level 2 •• From configuration to element

Identify the element and give its period, group and block: (a) $1s^2 2s^2 2p^6 3s^2 3p^4$; (b) $[\text{Ne}] 3s^1$; (c) $[\text{Ar}] 3d^6 4s^2$; (d) $[\text{Kr}] 5s^2$.

Answer

(a) 16 electrons: S, period 3, group 16, p block. (b) 11: Na, period 3, group 1, s block. (c) $18 + 8 = 26$: Fe, period 4, group 8, d block. (d) $36 + 2 = 38$: Sr, period 5, group 2, s block.

Q7. Level 2 •• From position to configuration

Write the shorthand configuration of the element located at: (a) period 3, group 15; (b) period 4, group 2; (c) period 4, group 17; (d) period 5, group 1. Name each element.

Answer

(a) $[\text{Ne}] 3s^2 3p^3$: phosphorus. (b) $[\text{Ar}] 4s^2$: calcium. (c) $[\text{Ar}] 3d^{10} 4s^2 4p^5$: bromine (the 3d subshell of period 4 is full after group 12). (d) $[\text{Kr}] 5s^1$: rubidium.

Q8. Level 2 •• Ground, excited or impossible?

Classify each configuration as ground state, excited state or impossible, and name the atom: (a) $1s^2 2s^2 2p^6 3s^1$; (b) $1s^2 2s^2 2p^5 3s^1$; (c) $1s^2 2s^3$; (d) $1s^2 2p^1$; (e) $1s^2 2s^2 2p^7$; (f) $1s^2 2s^2 2p^6 3d^1$.

Answer

(a) Ground state of Na (11 e^-). (b) Excited Ne (10 e^- , one electron promoted from 2p to 3s). (c) Impossible: an s subshell holds at most 2 electrons (Pauli). (d) Excited Li (3 e^- , 2p filled before 2s). (e) Impossible: p holds at most 6. (f) Excited Na (3d instead of 3s).

Q9. Level 3 ••• Ions and noble gases

Write the configuration of Na^+ , Mg^{2+} , O^{2-} , Cl^- and S^{2-} . Which noble gas does each resemble? Explain why these ions are the ones usually formed.

Answer

Na^+ (10 e^-), Mg^{2+} (10), O^{2-} (10): $1s^2 2s^2 2p^6 = [\text{Ne}]$. Cl^- (18), S^{2-} (18): $[\text{Ne}] 3s^2 3p^6 = [\text{Ar}]$. By losing or gaining a few valence electrons, each atom reaches the filled, very stable configuration of the nearest noble gas.

Q10. Level 3 ••• Iron and its ions

(a) Write the configuration of Fe ($Z = 26$) and draw the box diagram of its 3d and 4s subshells. How many unpaired electrons does it have? (b) Write the configuration of Fe^{2+} and Fe^{3+} . (c) Fe^{3+} is particularly stable. Suggest why.

Answer

(a) $[\text{Ar}] 3d^6 4s^2$; 3d: $\uparrow\downarrow\uparrow\uparrow\uparrow\uparrow$, 4s: $\uparrow\downarrow$: 4 unpaired electrons.
 (b) Remove 4s first: Fe^{2+} : $[\text{Ar}] 3d^6$; Fe^{3+} : $[\text{Ar}] 3d^5$.
 (c) $3d^5$ is a half-filled subshell: five parallel electrons, one per orbital, a particularly stable arrangement (the same reason as for chromium's exception).

Q11. Level 3 ••• Exceptions: chromium and copper

(a) Write the configuration predicted by Madelung's rule for Cr ($Z = 24$) and Cu ($Z = 29$). (b) The real configurations are $[\text{Ar}] 3d^5 4s^1$ and $[\text{Ar}] 3d^{10} 4s^1$. Explain. (c) How many unpaired electrons do Cr and Cu have?

Answer

- (a) Cr: $[\text{Ar}] 3d^4 4s^2$; Cu: $[\text{Ar}] 3d^9 4s^2$.
 (b) 3d and 4s are very close in energy. Moving one electron from 4s to 3d gives a half-filled (d^5) or completely filled (d^{10}) subshell, which is extra stable; the total energy is lower.
 (c) Cr: 3d $\uparrow\uparrow\uparrow\uparrow\uparrow$, 4s $\uparrow$: 6 unpaired. Cu: 3d full, 4s $\uparrow$: 1 unpaired.

Q12. Level 3 ••• Four quantum numbers

- (a) Give a possible set (n, ℓ, m_ℓ, m_s) for each of the five valence electrons of nitrogen in its ground state. (b) Which of these sets could describe an electron of a ground-state phosphorus atom? (i) $(3, 1, 0, +\frac{1}{2})$; (ii) $(3, 2, 0, +\frac{1}{2})$; (iii) $(2, 1, -1, -\frac{1}{2})$; (iv) $(3, 0, 0, 0)$.

Answer

- (a) $(2, 0, 0, +\frac{1}{2})$ and $(2, 0, 0, -\frac{1}{2})$ for $2s^2$; then, by Hund, $(2, 1, -1, +\frac{1}{2})$, $(2, 1, 0, +\frac{1}{2})$, $(2, 1, +1, +\frac{1}{2})$ for $2p^3$ (all $-\frac{1}{2}$ is equally correct).
 (b) P = $1s^2 2s^2 2p^6 3s^2 3p^3$. (i) yes, a 3p electron; (ii) no, 3d is empty in the ground state; (iii) yes, a core 2p electron; (iv) impossible, m_s must be $\pm\frac{1}{2}$.

Q13. Challenge ★ Why 2, 8, 8, 18, 18, 32?

Using Madelung's order and the fact that each period starts when a new ns subshell starts filling, list the subshells filled in each of the first six periods and deduce the number of elements in each.

Answer

Period 1: 1s (2). Period 2: 2s 2p ($2+6=8$). Period 3: 3s 3p (8; 3d comes after 4s). Period 4: 4s 3d 4p ($2+10+6=18$). Period 5: 5s 4d 5p (18). Period 6: 6s 4f 5d 6p ($2+14+10+6=32$). Period 7 is built the same way (7s 5f 6d 7p: 32).

Q14. Challenge ★ Magnetic atoms

Atoms with unpaired electrons are attracted by a magnet (paramagnetic); atoms with none are not (diamagnetic). Find the number of unpaired electrons of Mn ($Z=25$), Fe, Zn ($Z=30$), Cu and Zn^{2+} . Which are diamagnetic? Which ion of manganese, Mn^{2+} or Mn^{3+} , has the most unpaired electrons?

Answer

Mn $[\text{Ar}] 3d^5 4s^2$: 5. Fe $3d^6 4s^2$: 4. Zn $3d^{10} 4s^2$: 0. Cu $3d^{10} 4s^1$: 1. Zn^{2+} $[\text{Ar}] 3d^{10}$: 0.
 Diamagnetic: Zn and Zn^{2+} . Mn^{2+} $[\text{Ar}] 3d^5$ has 5 unpaired electrons, Mn^{3+} $[\text{Ar}] 3d^4$ has 4: Mn^{2+} has more.

Q15. Challenge ★ Beyond the known table

The last element of period 7 is oganesson, Og ($Z=118$). (a) Write the valence configuration of Og. (b) Using Madelung's rule, which subshells would period 8 fill, and in which order? (Remember that $\ell=4$ is called g.) (c) Predict the atomic number of the noble gas at the end of period 8.

Answer

- (a) $[\text{Rn}] 5f^{14} 6d^{10} 7s^2 7p^6$.
 (b) After 7p ($n+\ell=8$): 8s ($n+\ell=8$, larger n than 7p), then $n+\ell=9$ in order of increasing n : 5g, 6f, 7d, 8p.
 (c) $2+18+14+10+6=50$ elements, so the next noble gas would have $Z=118+50=168$. (Real super-heavy atoms are expected to deviate from this simple rule, but it is the Madelung prediction.)

Worksheet 5 · Atomic properties and periodicity

Data $h = 6.62 \times 10^{-34} \text{ J} \cdot \text{s}$ $c = 3.00 \times 10^8 \text{ m/s}$ $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$ $N_A = 6.022 \times 10^{23} / \text{mol}$
 $1 \text{ D} = 3.336 \times 10^{-30} \text{ C} \cdot \text{m}$. Use Tables 5-1 to 5-4 of the handout (radii, ionisation energies, bond energies, Pauling electronegativities). Calculator allowed.

Q1. Level 1 • Definitions

Define: atomic radius, first ionisation energy, polarisability, electronegativity. For each, say whether it is a property of an isolated atom or of an atom in a bond.

Answer

Atomic radius: distance from the nucleus to the outer edge of the electron cloud (radius of the highest occupied AO); measured on atoms in molecules or solids. First ionisation energy: minimum energy to remove the outermost electron from an atom in the gas phase, $\text{X(g)} \rightarrow \text{X}^+(\text{g}) + \text{e}^-$; isolated atom. Polarisability: how easily the electron cloud of an atom is distorted by an electric field; isolated atom. Electronegativity: ability of an atom *in a bond* to attract the shared electrons.

Q2. Level 1 • Average atomic mass of chlorine

Chlorine is 75.77 % ^{35}Cl (34.97 amu) and 24.23 % ^{37}Cl (36.97 amu). Compute its average atomic mass and its molar mass.

Answer

$$\bar{m} = 0.7577 \times 34.97 + 0.2423 \times 36.97 = 26.50 + 8.96 = 35.45 \text{ amu}; M = 35.45 \text{ g mol}^{-1}.$$

Q3. Level 1 • Ranking radii

Using Table 5-1, rank Na, Mg, K and Cl by increasing atomic radius and state the two trends that explain the order.

Answer

Cl (99) < Mg (160) < Na (191) < K (235 pm). The radius decreases across a period (Na > Mg > Cl, period 3) and increases down a group (K > Na, group 1).

Q4. Level 1 • Units of ionisation energy

The ionisation energy of sodium is 496 kJ mol^{-1} . Convert it to joules per atom and to electronvolts per atom. Show that 1 eV per atom corresponds to 96.4 kJ mol^{-1} .

Answer

$$496 \times 10^3 / 6.022 \times 10^{23} = 8.24 \times 10^{-19} \text{ J per atom} = 8.24 \times 10^{-19} / 1.6 \times 10^{-19} = 5.15 \text{ eV}.$$

$$1 \text{ eV} \times N_A = 1.6 \times 10^{-19} \times 6.022 \times 10^{23} = 9.64 \times 10^4 \text{ J mol}^{-1} = 96.4 \text{ kJ mol}^{-1} \text{ (96.5 with the exact value of } e).$$

Q5. Level 2 •• Mass versus atomic number

- (a) Compute the average atomic mass of potassium: ^{39}K 38.96 amu, 93.26 %; ^{41}K 40.96 amu, 6.73 %.
 (b) Argon (39.95 amu) has $Z = 18$ and potassium $Z = 19$. What problem would appear if elements were ordered by mass? How is it solved today?

Answer

- (a) $0.9326 \times 38.96 + 0.0673 \times 40.96 = 36.33 + 2.76 = 39.09 \text{ amu}$.
 (b) By mass K (39.09) would come before Ar (39.95): K would fall under the noble gases and Ar under the alkali metals, against their chemistry. The table is ordered by Z (number of protons),

which puts Ar in group 18 and K in group 1.

Q6. Level 2 •• Explaining the radius trends

Explain, using the valence shell n and the effective nuclear charge, why the radius decreases from Li (157 pm) to F (64 pm) and increases from F to I (133 pm). Why can Ne (154 pm) not be compared with F?

Answer

Li to F: same valence shell ($n = 2$), but Z_{eff} increases because each added electron is in the same shell and screens the added proton poorly; the cloud is pulled in. F to I: n goes from 2 to 5, the valence orbitals are much larger ($r \propto n^2$), while Z_{eff} barely changes. Ne's value is a van der Waals radius (non-bonded atoms), always much larger than a covalent radius.

Q7. Level 2 •• An isoelectronic series

Rank O^{2-} , F^- , Na^+ and Mg^{2+} by increasing radius. Justify. Compare also the radius of Na^+ with that of Na, and of Cl^- with that of Cl.

Answer

All have 10 electrons ($1s^2 2s^2 2p^6$) but 8, 9, 11 and 12 protons. More protons for the same electrons pull the cloud in: $\text{Mg}^{2+} < \text{Na}^+ < \text{F}^- < \text{O}^{2-}$. Na^+ is much smaller than Na (it loses its whole $n = 3$ shell); Cl^- is larger than Cl (one more electron, more repulsion, same nucleus).

Q8. Level 2 •• The ionisation energies of period 2

The values for period 2 are: Li 520, Be 900, B 801, C 1086, N 1402, O 1314, F 1681, Ne 2081 kJ mol^{-1} .

(a) State the general trend and explain it. (b) Explain the two exceptions with box diagrams.

Answer

(a) E_I increases across the period: Z_{eff} increases and the radius decreases, so the electron is held more strongly.

(b) $\text{Be} > \text{B}$: B loses its single 2p electron ($\uparrow\downarrow \uparrow \square \square$), higher in energy and screened by 2s, easier to remove than a 2s electron of Be ($\uparrow\downarrow \square \square \square$). $\text{N} > \text{O}$: N has a half-filled 2p ($\uparrow\downarrow \uparrow \uparrow \uparrow$); O has one pair ($\uparrow\downarrow \uparrow \uparrow \uparrow$), and the repulsion between the two paired electrons makes one of them easier to remove.

Q9. Level 3 ••• Ionising light

(a) Compute the longest wavelength of a photon able to ionise a sodium atom (496 kJ mol^{-1}), and the one for helium (2372 kJ mol^{-1}). In which region of the spectrum are they? (b) In a PES experiment, UV photons of 21.2 eV hit sodium atoms. What is the kinetic energy of the ejected valence electrons?

Answer

(a) Na: $E = 8.24 \times 10^{-19} \text{ J}$ (Q4), $\lambda = hc/E = 6.62 \times 10^{-34} \times 3.00 \times 10^8 / 8.24 \times 10^{-19} = 2.41 \times 10^{-7} \text{ m} = 241 \text{ nm}$ (UV). He: $E = 2372 \times 10^3 / 6.022 \times 10^{23} = 3.94 \times 10^{-18} \text{ J}$, $\lambda = 50.4 \text{ nm}$ (far UV).

(b) $E_k = h\nu - E_I = 21.2 - 5.15 = 16.0 \text{ eV}$.

Q10. Level 3 ••• Successive ionisation energies

An element of period 3 has successive ionisation energies 738, 1451, 7733, 10 543 kJ mol^{-1} . (a) Identify it and justify. (b) Why does each ionisation energy exceed the previous one? (c) Which ion does this element form in its compounds?

Answer

- (a) The huge jump comes between the 2nd and the 3rd (1451 $\rightarrow$ 7733): 2 electrons are easy to remove, the 3rd is a core electron (2p). Two valence electrons in period 3: magnesium, [Ne] 3s².
 (b) Each electron is removed from an increasingly positive ion: fewer electrons for the same nuclear charge, less repulsion, the remaining electrons are held more tightly.
 (c) Mg²⁺ ([Ne]).

Q11. Level 3 ●●● Abundance from the average mass

Boron has two isotopes, ¹⁰B (10.013 amu) and ¹¹B (11.009 amu), and an average atomic mass of 10.81 amu. Compute the natural abundance of each isotope.

Answer

With x the fraction of ¹⁰B: $10.013x + 11.009(1 - x) = 10.81$, so $x = (11.009 - 10.81)/(11.009 - 10.013) = 0.199/0.996 = 0.200$: 20.0 % ¹⁰B and 80.0 % ¹¹B.

Q12. Level 3 ●●● How polar is H–Cl?

The dipole moment of HCl is 1.08 D and its bond length 127 pm. (a) Compute the partial charge δ (as a fraction of e) carried by each atom. Which atom is δ^- ? (b) Using Table 5-4, rank H–F, H–Cl, H–Br and H–I by increasing polarity.

Answer

- (a) $\delta e = \mu/d = 1.08 \times 3.336 \times 10^{-30} / 127 \times 10^{-12} = 2.84 \times 10^{-20}$ C, so $\delta = 2.84 \times 10^{-20} / 1.6 \times 10^{-19} = 0.18$. Cl is more electronegative (3.16 vs 2.2): Cl is δ^- , H is δ^+ . The bond is about 18 % ionic.
 (b) $\Delta\chi$: H–I 2.66 – 2.2 = 0.46 < H–Br 0.76 < H–Cl 0.96 < H–F 1.78.

Q13. Challenge ★ Pauling electronegativity from bond energies

Use Table 5-3 (H–H 436, F–F 158, H–F 567, I–I 151, H–I 299 kJ mol^{–1}), the formula $|\chi_A - \chi_B| = 0.102\sqrt{\Delta}$ with $\Delta = D_{AB} - \sqrt{D_{AA}D_{BB}}$, and $\chi_H = 2.20$. (a) Compute χ_F . (b) Compute χ_I and compare with Table 5-4 (2.66). (c) Why is the H–F bond much stronger than the average of H–H and F–F?

Answer

- (a) $\sqrt{436 \times 158} = 262.5$; $\Delta = 567 - 262.5 = 304.5$; $0.102\sqrt{304.5} = 1.78$; F is more electronegative than H: $\chi_F = 2.20 + 1.78 = 3.98$ (matches Table 5-4).
 (b) $\sqrt{436 \times 151} = 256.6$; $\Delta = 299 - 256.6 = 42.4$; $0.102\sqrt{42.4} = 0.66$; $\chi_I = 2.20 + 0.66 = 2.86$, 0.2 above the table. Pauling's values are fitted over many molecules; one pair only gives an estimate, and small Δ values are very sensitive to errors.
 (c) In the polar H–F bond the partial charges δ^+ and δ^- attract each other: this ionic contribution adds to the covalent bond energy. The more different the electronegativities, the larger this extra energy Δ .

Q14. Challenge ★ The Allred–Rochow scale

(a) Using Slater's rules (Unit 5, "Going further"), show that $Z_{\text{eff}} = 6.1$ for a valence electron of Cl. (b) With $\chi = 0.359 Z_{\text{eff}}/r^2 + 0.744$ (r in Å) and the covalent radius of Cl (99 pm), compute $\chi_{\text{AR}}(\text{Cl})$ and compare with Pauling's 3.16. (c) Without calculation, explain why F has a higher electronegativity than Cl on this scale.

Answer

- (a) Cl = 1s² 2s² 2p⁶ 3s² 3p⁵. For one 3p electron: 6 other $n = 3$ electrons $\times 0.35 = 2.10$; 8 electrons of $n = 2$ $\times 0.85 = 6.80$; 2 electrons of $n = 1$ $\times 1.00 = 2.00$; $\sigma = 10.9$ and $Z_{\text{eff}} = 17 - 10.9 = 6.1$.
 (b) $r = 0.99$ Å: $\chi = 0.359 \times 6.1/0.99^2 + 0.744 = 2.23 + 0.74 = 2.98$. The two scales agree within about 0.2 (they are built on different quantities: a force for Allred–Rochow, bond energies for

Pauling).

(c) F has a similar Z_{eff} (5.2 by Slater's rules) but a much smaller radius (64 pm): its valence shell is $n = 2$. Z_{eff}/r^2 is larger, so the bonding electrons are attracted more strongly.

Q15. Challenge ★ **Synthesis: an unknown element**

Element X is in period 3. Its successive ionisation energies are 578, 1817, 2745 and 11 577 kJ mol⁻¹.

(a) Identify X and write its configuration with the box diagram of its valence shell. (b) Explain why the first ionisation energy of X is lower than that of Mg (738), although X has one more proton. (c) Compare the radius, the electronegativity and the polarisability of X with those of Mg and Si, using the tables. (d) Which ion does X form? Is it isoelectronic with a noble gas?

Answer

(a) The jump is after the 3rd electron (2745 → 11 577): 3 valence electrons in period 3: aluminium, [Ne] 3s²3p¹, $\uparrow\downarrow$ $\uparrow$ $\square$ $\square$.

(b) Al loses its single 3p electron, which is higher in energy and partly screened by the 3s pair; Mg loses a 3s electron, more penetrating and more tightly held (same anomaly as Be/B).

(c) Radius: Mg 160 > Al 143 > Si 118 pm (decreases across the period). Electronegativity: Mg 1.31 < Al 1.61 < Si 1.90 (increases). Polarisability follows the radius: Mg > Al > Si.

(d) Al³⁺, [Ne]: isoelectronic with neon.